

Spring 2026. Physics 510.

Quantum Theory II.

Homework # 2.

Due to February 16th, 2026.

Superconductivity. Desai book, chapter 39. Sections 39.1-39.3 (feel free to move further)

- a) Read about a Fermi surface and quasiparticles as its excitations.
- b) Discuss the original interaction term 39.1. What assumptions, simplifications etc. did we use?
- c) Do all calculations to derive formula 39.9 from 39.3.
- d) Separately discuss the key approximation 39.5.
- e) Now go step by step to derive formula 39.33 and then, with further simplifications, formula 39.37.
- f) Analyze both $\Delta = 0$ (normal), and $\Delta \neq 0$ (superconducting) cases.
- g) Is it possible to explain the superconductivity phenomenon using any version of a perturbation theory? Why? (Hint: look at the gap equation 39.57).